Experiment 6: Workflow Automation using Deluge

1. AIM

To study and implement workflow automation using Deluge scripting in Zoho Creator by configuring triggers, applying conditions, executing automated actions, and reducing manual effort in application processes.

2. THEORY

Workflow Automation

Workflow automation is the process of executing predefined actions automatically when a particular event occurs or a specified condition is satisfied. It helps reduce repetitive tasks, improve data accuracy, and maintain consistency in business processes.

Zoho Creator provides workflow automation features that allow developers to perform actions when records are created, edited, deleted, or when other configured events occur.

Deluge Scripting

Deluge (Data Enriched Language for the Universal Grid Environment) is Zoho's scripting language used to implement business logic and automate application processes. It supports variables, conditions, loops, record operations, notifications, and integrations.

Deluge scripts can be configured in workflows to validate information, calculate values, update records, send notifications, and execute other automated tasks.

Components of Workflow Automation

1. **Trigger:** Defines the event that starts the workflow, such as adding or editing a record.
2. **Condition:** Specifies the criteria that must be satisfied before an action executes.
3. **Action:** Defines the task performed by the workflow, such as updating a field or sending a notification.
4. **Deluge Script:** Contains the logic used to execute custom actions.
5. **Execution:** Occurs when the configured trigger and conditions are met.

Applications of Workflow Automation

- Sending automated email notifications.
- Updating record statuses.
- Validating user-entered data.

- Calculating values automatically.
- Assigning tasks to users.
- Generating alerts and reminders.
- Integrating with other applications.

3. PROCEDURE

Step 1: Open Zoho Creator

1. Log in to Zoho Creator.
2. Open the required application.
3. Navigate to the application builder.
4. Select the **Workflow** section.

Step 2: Create a Workflow

1. Click the option to create a new workflow.
2. Enter a suitable workflow name.
3. Select the form on which the workflow must operate.
4. Choose the required trigger, such as **On Create** or **On Update**, depending on the task.

Step 3: Configure the Trigger

Select the event that should initiate the workflow. For example, when a new vehicle service request is submitted, the workflow can execute automatically.

Step 4: Add Conditions

Configure conditions to determine when the workflow should execute. For example, the workflow may run only when the service type is Breakdown Repair.

Step 5: Write a Deluge Script

The following example displays an informational message when a vehicle service request is submitted.

```
None
info "Vehicle service request submitted successfully.";
```

Add this script to an appropriate form workflow action supported by the current Creator interface.

Step 6: Create a Status-Based Workflow

For a vehicle service application, a workflow can update a request's status based on the selected service information.

Example Deluge logic:

```
None
if(input.Service_type == "Breakdown Repair")
{
    info "Priority service request received.";
}
```

This example checks whether the selected service type is Breakdown Repair and displays a message. It does not automatically change the priority or send a notification.

Step 7: Save and Test the Workflow

1. Save the workflow configuration.
2. Open the application form.
3. Submit a test record that satisfies the configured trigger and conditions.
4. Observe whether the workflow executes.
5. Verify the result and correct any errors if necessary.

4. EXAMPLE APPLICATION

In the Vehicle Service & Repair Request Management application, workflow automation can be used for the following tasks:

Event	Automated Action
New service request submitted	Display confirmation or send an acknowledgement email
Service status updated	Notify the vehicle owner of the status change
Repair marked completed	Update the repair completion status
Preferred service date approaching	Send a reminder using a suitably configured schedule
Breakdown repair selected	Display a priority message or trigger a configured assignment action

These actions can be implemented using workflow configuration and Deluge scripts, depending on the application's field names and available permissions.

5. RESULT

Workflow automation using Deluge in Zoho Creator was studied, and example scripts were developed to demonstrate event-based execution and conditional logic. The experiment illustrated how workflows can automate repetitive tasks, execute actions based on conditions, and improve the efficiency of low-code applications.

Thus, the experiment demonstrated the use of Deluge scripting for implementing business logic and automating application workflows.



